

Development of Smart Antenna Array Signal Processing Algorithm for Anti-Jam GPS Receiver

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Abstract— GPS guidance uses low power signals from satellites which are 11000 NMI away. The satellite transmitter power is modest nearly 10 W orders of magnitude. Neither satellite nor receivers have the luxury of very high antenna gain since both entities have significant field of view requirements. These factors result in a very low power density incident on a GPS receiver antenna. The signal received is generally 165 dB down than the thermal noise level. Such signals are notoriously easy to jam either by intentional noise sources (Jammer) or unintentionally from harmonics of broadcasting stations or other out of band sources. Here in this paper we will show how a nulling antenna or controlled reception pattern antenna with adaptive spatial filtering technique efficiently mitigate the intentional and non intentional interferences. A beamforming antenna array is a set of antennas whose outputs are weighted by complex values and combined to form the array output. The effect of the complex valued weights is to steer main lobes of the array pattern to desired directions. These directions may be unknown and so the antenna weights must be adjusted adaptively until some measure of array performance is improved, indicating proper lobe or null placement. An adaptive algorithm to adjust the complex weights of an antenna array is presented that nulls high power signals while allowing reception of GPS signals as long as the signals arrive from different directions.

1. INTRODUCTION

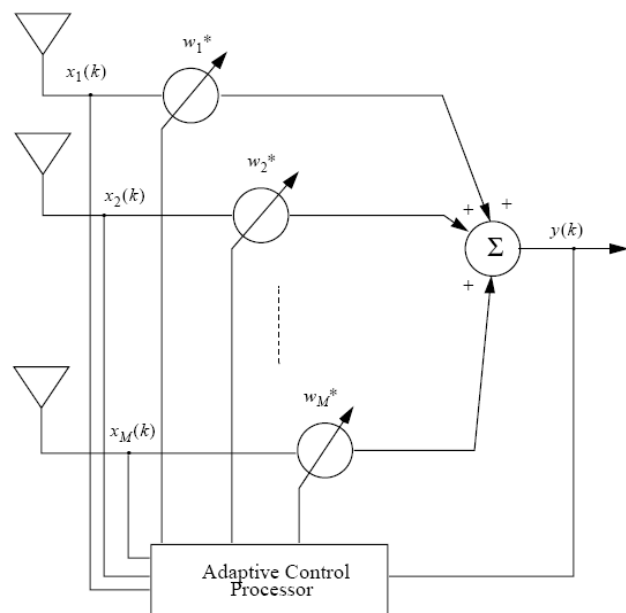
The GPS signals are spread spectrum modulated and have very low average power, on the order of background thermal noise. Simulations of the adaptive algorithm minimize the output power of the array to within 5 dB of the background noise level. The technique which will be adopted to optimize the weight values is steepest descent algorithm and implements an efficient, exact gradient calculation. With M antennas in the array, only $M-1$ weights are adjustable. It appears that $M-1$ adjustable antenna weights can null $M-1$ unwanted signals (jammers). However, in the course of the algorithm development, a few configurations of antennas and jammer arrival directions were found where this is not true. Even when the jammer arrival directions are known, certain configurations are mathematically impossible to cancel out the intentional interferences.

The basic requirements of an adaptive Nulling antenna array are that it should provide enough gain margin to the user (or users) to satisfy the link calculations and at the same time suppressing the interfering signals. To achieve this, the elements of the antenna should have enough gain individually and should be physically configured in such a way as to be able simultaneously to point pattern nulls in the direction of interfering sources. The algorithm is the most crucial in steering the main beam in the direction of the SOI (signal of interest). Incoming signal to the array are of three types 1. GPS Signal 2. Jammer signal 3. Noise The GPS signal and Noise are considered to have low power of the order of background noise level, while Jammers have assumed to have much higher power level. The strategy behind this adaptive nulling aiming on reducing the array output Jammer power to a level comparable to the output GPS signal power, so that the later can be detected with spread spectrum technology.

2. ANTI-JAM ECCM TECHNIQUES

Basic to any AJ receiver design are the abilities to detect the presence of jamming and characterize jamming features. Detection of jamming can be accomplished by comparing signal levels to the noise floor in the receiver, prior to the code correlators. At this stage in the receiver, the GPS signal amplitude is well below the noise floor. Detection of significant signal amplitude at this stage is indicative of the presence of jamming or other interference. Automatic gain control (AGC) measurements can characterize the amplitude of the jamming signal. A discrete Fourier transform

(DFT) provides an effective spectral characterization. Once the presence of jamming is established, a GPS receiver with AJ features can begin to adapt to the jamming environment within which it finds itself. The GPS receiver's next layers of defense, that are aimed at actually reducing the level of jamming contaminating signal processing functions, are embodied in various levels and types of filtering. At the first level, GPS receivers require fixed bandwidth front-end RF filters that pass the L-band frequencies of the spread spectrum satellite signals while simultaneously rejecting out-of-band jamming/interference. Out-of-band jamming would probably not be intended to inhibit GPS users but could incidentally disable receivers if not eliminated early in the receive chain. At the next level, military receivers should have intermediate frequency (IF) adaptive notch filters to counter narrowband noise and CW tone jammers. As described in the previous section, these jamming signals can be located in the frequency domain via DFT measurements performed by the jammer detection process. Once detected and isolated in frequency, various spectral and temporal signal processing techniques can be employed to implement a band rejection (or notch) filter that is adaptively tuned to the narrow frequency interval occupied by the jammer. Such notch filtering techniques can provide 15–30 dB of jamming suppression with only minor distortion/attenuation of the GPS signal, since only a small fraction of the signal's spread bandwidth is affected by the filter. Antenna-based ECCM techniques fall into the basic categories of antenna switching, polarization nulling, and adaptive array processing. Adaptive array processing includes the techniques of beam steering toward satellites, null steering (AKA spatial filtering) toward jammers, and combinations thereof aimed at optimizing the overall signal-to jammer power ratio (SJR). Antenna switching technique employs multiple antennas with narrower main beams than the hemispheric coverage typical of most GPS antennas. In a jamming environment, antennas are selected that have the best combination of gain in the satellite's direction and low side lobes in the jammer's direction — SJR is then maximized for the available set of antennas. However, jammer suppression is relatively marginal for this technique (~ 10 dB) and installation of multiple directional antenna apertures is impractical for some host platforms (especially for Guided Weapons). **Adaptive array processing** techniques attempt to remedy the drawbacks of the antenna switching concept by utilizing arrays of small antenna elements rather than full directional apertures. This makes installation on even small weapon platforms practical. Furthermore, implementation of phased-array processing is the key to significant multiple jammer suppression that can substantially restore GPS receiver operation in extensive jamming environments. Adaptive arrays provide phase control to each element of their multiple element antennas. In the absence of jamming, the individual phases are adjusted



(a)

Anti-Jam Signal Processing Techniques	Jammer Rejection Performance	Cost/Complexity	Implementation
Pre-correlation Adaptive Nulling Antenna Processing	30-50 dB	High	Analog/Digital
Pre-correlation Temporal/Spectral Processing	20-30 dB	Very Low	Digital
Post-correlation Processing	10-15 dB	Low	Digital Hardware/Software

(b)

Figure 1: Adaptive antenna array with adaptive control processor, (b) comparison of adaptive signal processing technique with respect to performance cost & implementation.

and combined to create hemispheric satellite coverage by the resulting antenna reception pattern. In the presence of jamming, the phases are adjusted to steer high gain beams in the direction of satellites and/or low gain nulls in the direction of jammers. If null steering is implemented exclusively, the adaptive array is referred to as a controlled reception pattern antenna (or CRPA). The number of nulls a CRPA can generate and steer is dictated by the number of antenna elements in the array — the number of nulls being one less than the number of elements.

3. ADAPTIVE ANTENNA ANTI-JAM SIGNAL PROCESSING TECHNIQUE

An adaptive antenna controls its radiation pattern by mixing the received RF signals from multiple antenna elements with controlled complex weights (phase shifters and attenuators) and by summing the resulting signals. The radiation pattern of the final combined signal depends on the radiation pattern of each antenna element, on the geometry of the location of the antenna elements, and on the controlled antenna weights. Jammer suppression is accomplished by generating and using antenna weights so that the resulting radiation pattern minimizes power reception in the direction of the jammer. This system is referred to as a null steering (NS) antenna. Typically, the generation of antenna weights in adaptive NS systems is accomplished by an optimization algorithm which continuously tries to minimize the power of the interference RF signal at the output of the antenna combiner subjected to a suitable constraint (one fixed weight) to ensure signal reception.

Adaptive null steering antennas require, at a minimum, one controlled complex weight for each null in the radiation pattern in addition to one for the first antenna element. For example, we need three antenna elements to suppress two jammers radiating from different directions. These systems are very powerful, in that (1) they do not require the jammer to have distinct signal characteristics compared to the GPS satellite signal and (2) they can provide high processing gain, i.e., improvement in the signal-to-noise ratio (SNR) at the receiver input, often exceeding 30 dB. Unfortunately, adaptive antenna systems are prone to nulling out GPS signals if the direction of arrival of the jamming signal is close to that of the GPS. Also, adaptive antenna systems are very expensive compared to the cost of alternate anti-jamming systems, or even to the cost of the receiver unit itself.

The above two limitations of adaptive RF nulling antenna can be removed by using an optimization criterion (i.e., cost function) which penalizes nulling of a desired signal, such as minimizing J/S where J is the jammer power and S is desired signal, or directly maximizing signal-to-interference-plus-noise power ratio (SINR). The choice of optimization criterion along with the choice of the optimization algorithm determines the performance of the adaptive antenna. The limitation associated with the choice of RF analog implementation for the antenna electronics can be easily removed by employing digital signal processing to effectively implement the antenna gain and phase control and combiner functions.

4. DIRECTION OF ARRIVAL (DOA) ESTIMATION FOR ADAPTIVE PROCESSOR

In array signal processing, the estimation of the direction of arrival angle (DOA) from multiple sources plays an important role, because we need to feed the desired and jamming signal direction to adaptive processor so that it can produce null and steer main lobe towards the desired signal. Usually we employ multiple antenna array elements, and their array signal processing can increase the capacity and throughputs of the system significantly. Estimation of the DOA's of incoming signals can be used to localize the signal sources and characterize them.

Among the various mathematical realization DOA estimation algorithm, like Capon's beamformer, conventional beamformer, MUSIC algorithm, ESPRIT algorithm we choose the last one with some modification in mathematical realization for DOA estimation for Adaptive processor.

5. C-ESPRIT ALGORITHM

Estimation of signal parameters via rotational invariance techniques (ESPRIT) is a signal subspace technique. ESPRIT is a computationally efficient and robust method of DOA estimation. It considers two identical sub arrays consisting of the same number of antenna elements, and each matched pair of elements is called a doublet with an identical displacement vector. In the ESPRIT algorithm, the number of a doublet depends on overlapping between sub arrays. Considering a uniform linear array (ULA) consisting of M elements, then the number of doublets in the case of no overlapping is equal to half the number of elements $M = 2m$, where m is the number of doublets. But if maximum overlapping occurs between the two sub arrays, then the number of doublets becomes $m = M - 1$, compared to the MUSIC algorithm, ESPRIT does not require an

exhaustive search through all possible steering vectors to estimate the DOA. Moreover, ESPRIT reduces computational complexity and storage requirements, which makes real time implementation possible. C-ESPRIT algorithm can provide a more precise DOA estimation and can detect more signals than the well-known classical subspace methods, MUSIC and ESPRIT, for the 1-D and 2-D DOA. The complexity is the same as that of ESPRIT, since the proposed algorithm uses the same array geometry and sub array processing as ESPRIT. The main differences between the proposed and ESPRIT algorithms are as follows: (1) in the proposed algorithm, the number of overlapping array elements between the two sub arrays is equal to M , while in ESPRIT, the maximum number of overlapping elements is $M-1$, where M denotes the total number of array elements, and (2) the proposed algorithm employs the conjugate of rotation matrix (CRM) Φ^* , while ESPRIT uses Φ with no conjugate for the second sub array geometry.

Considering a uniformly linear array of M elements and assuming K non-coherent and narrowband one dimensional signals are received at uniform linear array(ULA) with different DOA's $\theta_1, \theta_2, \dots, \theta_k$, the $M \times 1$ received signal vector can be written as $X(t) = \sum_{k=1}^K a(\theta_k) S_k(t) + n(t)$. Where $S_k(t)$ represents the signal from the k th source with DOA equal to θ_k , $a(\theta_k)$ which denotes the $M \times 1$ array response vector and $n(t)$ is the $M \times 1$ AWGN vector with each component of mean zero and variance equal to σ^2 . The array response vector can be written as $a(\theta_k) = (1, Z_k, Z_k^2, \dots, Z_k^{M-1})^T$ and $Z_k = \exp(-j \frac{2\pi d \cos(\theta_k)}{\lambda})$ where λ the wave length, d is spacing between elements. $M \times K$ array response matrix and $K \times 1$ signal vector can be written as $A(\theta) = [a(\theta_1), a(\theta_2), \dots, a(\theta_k)]$, and $S(t) = (S_1(t), S_2(t), S_3(t), \dots, S_k(t))^T$ respectively. C-SPRIT algorithm can estimate either azimuth DOA φ_k or the joint azimuth and elevation DOA (φ_k, θ_k) . For the sub-array1 we form column vector $(y_1, y_2, \dots, y_M)^T$. In fact two separate sub array do not exist they are fully overlapped each sub array processing in a C-SPRIT uses the maximum number of array elements equal to M where as each sub-array processing in ESPRIT can have only $M/2$ elements for the doublet case or $M-1$ at most, for the maximum overlap case, this is the reason why C-SPRIT have better SNR. 1st element in the array is treated as the reference with respect to the other elements. In this case the $M \times 1$ input vector to sub array can be written as

$$Y_1(t) = A(\theta)S(t) + n_1(t) \quad (1)$$

$$A = A(\theta) = \begin{bmatrix} 1 & \dots & 1 \\ Z_1 & \dots & Z_k \\ Z_1^2 & & \\ \vdots & & \\ Z_1^{M-2} & \dots & Z_k^{M-2} \end{bmatrix}, \quad Y_1(t) = [y_1(t), y_2(t), \dots, y_M(t)]^T \text{ and } n(t) = [n_1(t), n_2(t), \dots, n_M(t)]^T.$$

$A(\theta)$ is the array response matrix with dimension $M \times K$, $S(t)$ supposed to be narrowband signal vector with dimension $K \times 1$, $n_1(t)$ is $M \times 1$ AWGN vector whose component are zero mean and variance σ^2 . Input vectors to sub array 2 is denoted as $Y_2(t) = [y_2^*, y_1, \dots, y_{M-1}]^T$, if we represent it in terms of $A(\theta)$, $S(t)$, ϕ^* it becomes

$$Y_2(t) = \begin{pmatrix} \sum_{k=1}^K S_k Z_k^* + n_2^* \\ \sum_{k=1}^K S_k + n_1 \\ \sum_{k=1}^K S_k Z_k + n_2 \\ \vdots \\ \sum_{k=1}^K S_k Z_k^{M-2} + n_{M-1} \end{pmatrix} = A(\theta)\phi^* S(t) + n_2(t). \quad (2)$$

For one dimensional signal $\phi^* = \text{diag}(Z_1^*, Z_2^*, \dots, Z_k^*)$. Y_1 & Y_2 the total output vector $Z(t)$ can be written as,

$$Z(t) = \begin{bmatrix} Y_1(t) \\ Y_2(t) \end{bmatrix} = BS(t) + n(t) \quad (3)$$

$$B = \begin{bmatrix} A \\ A\phi^* \end{bmatrix}, \quad n(t) = \begin{bmatrix} n_1(t) \\ n_2(t) \end{bmatrix}.$$

B is $2M * K$ matrix and both $Z(t)$ and $n(t)$ are $2M * 1$ vectors forming $2M * 2M$ covariance matrix $Z(t)$.

$$R_{zz} = BR_s B^H + \sigma^2 I \quad (4)$$

$R_s = E[S(t)S(t)^H]$ is a $K * K$ signal covariance matrix. As the signals are uncorrelated and $2M-K$ smallest eigen values of R_{zz} are equal to σ^2 . The K eigen-vectors corresponds to K largest eigen values can be written in a $2M * K$ matrix $F_s = [e_1, e_2, \dots, e_k]$. The range space of F_s is equal to B i.e., $R(F_s) = R(B)$, thus a nonsingular matrix T exist such that

$$\begin{aligned} F_s &= BT; \\ F_s &= \begin{bmatrix} F_0 \\ F_1 \end{bmatrix} = \begin{bmatrix} AT \\ A\phi^* T \end{bmatrix} \end{aligned} \quad (5)$$

$\phi^* = T\psi T^{-1}$ Where the Eigen-values of matrix ψ are equal to the diagonal elements of ϕ^* and the columns S of T are the eigenvectors of ψ . To find the DOA steps to be followed may be summarized as

1) Obtain the estimate of R_{zz} .

2) Apply Eigen-value decomposition of R_{zz} . $R_{zz} = F\Lambda_{zz}F^H$.

$$\Lambda_{zz} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_{2M}) \text{ and } F = [e_1, e_2, \dots, e_{2M}].$$

3) Use multiplicity of K of the smallest Eigen value to estimate the number of signals as $\hat{K} = 2M - K$.

4) Estimate the signal subspace $\hat{F}_s$ using the eigenvector correspond to the largest K Eigen values and decompose F_s into two $M * K$ sub matrices F_0 & F_1 .

5) Apply Eigen value decomposition on the matrix formed as $G = \begin{bmatrix} \hat{F}_0^H \\ \hat{F}_1^H \end{bmatrix} [\hat{F}_0 \hat{F}_1] = F\Lambda_G F^H$.

$$\text{Partition } F \text{ into } K * K \text{ sub matrices as } F = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix}.$$

6) Calculate the Eigen value λ_k of ψ When $\psi = -F_{12}F_{22}^{-1}$ then estimate k th diagonal element ϕ_k^* ; $\phi_k^* = \lambda_k$.

7) Estimate azimuth DOA $\phi_k^* = \cos^{-1} \left[\frac{\arg(\hat{\phi}_k^*)}{2\pi d/\lambda} \right]$. For realization of joint azimuth & elevation 2D DOA estimation with C-SPRIT we need to collect data from the array elements and Z_K will be function of both elevation DOA θ_k and azimuth DOA ϕ_k and $Z_K = \exp \left(-j \frac{2\pi d \cos \hat{\phi}_k \sin \theta_k}{\lambda} \right)$ and $\hat{\phi}_k^* = \cos^{-1} \left[\frac{\arg(\phi_k^*)}{2\pi d \sin \theta_k / \pi} \right]$.

6. SIMULATION RESULT

Assumption is made that $K = 3$, signals are from BPSK sources. ULA of 6 elements separated by distance equal to half wave length of incoming signal employed. The no of data samples per trial at each element output is $N = 200$ and no of total independent trial is 2500. Figure shows the histogram plot of azimuth DOA estimation for DOA at 82° , 90° , 98° and SNR = $[2, 2, 2]$ dB. It becomes clear that proposed algorithm gives the most accurate DOA estimation most of the time all peaks are observed at around 82° , 90° , and 98° .

7. LMS ALGORITHM

This algorithm is based on method of steepest decent. Changes in weight vectors are made along the direction of the estimated gradient vector.

Accordingly,

$$W(j+1) = W(j) + k_s \hat{\nabla}(j) \quad (6)$$

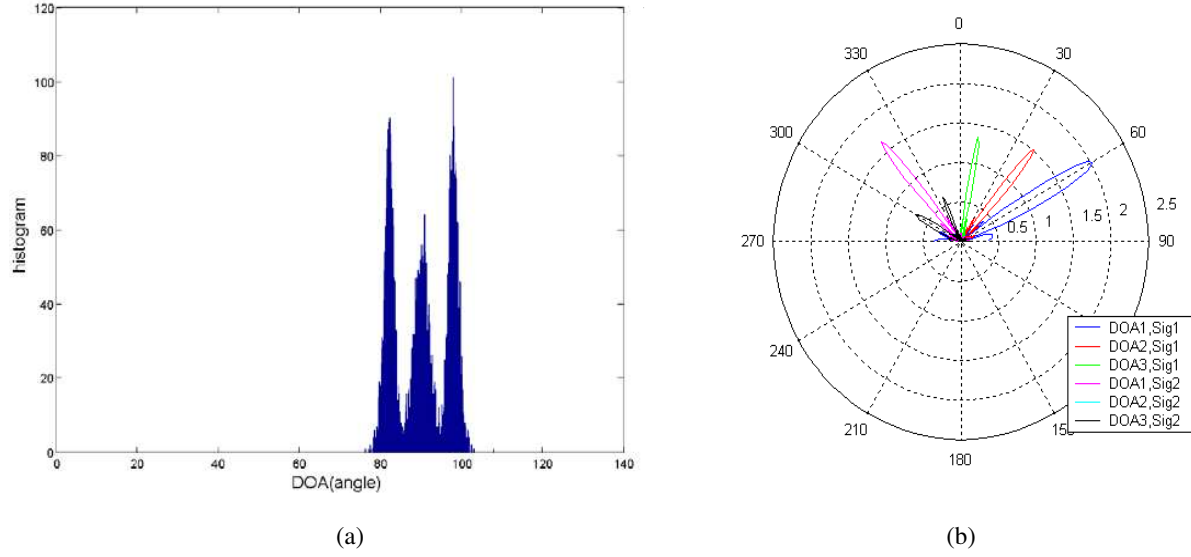


Figure 2: (a) Histogram of DOA estimations for $K = 3$ sources of DOA at $[82^\circ, 90^\circ, 98^\circ]$, $\text{SNR} = [2, 2, 2]$ dB, and $M = 6$ elements for C-SPRIT, (b) adaptive antenna scanning two signals from 60° (with two multipath form 40° & from 10°) & -40° (with two multipath form -60° & from -20°).

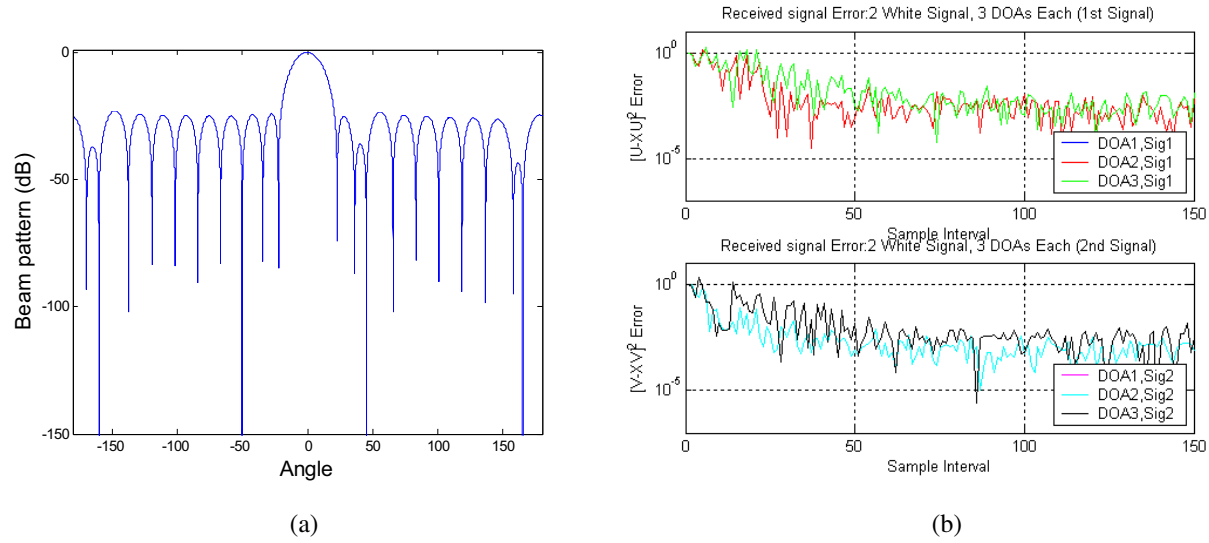


Figure 3: (a) Multiple null steering desired signal coming from -20° to 20° Jammers are from $45^\circ, 165^\circ, -50^\circ$ and -160° (4 Jammers), (b) LMS algorithm converging around 49–50 iteration for all signals.

where $W(j)$ = weight vector before adaptation, $W(j+1)$ = weight vector after adaptation, k_s = scalar constant controlling rate of convergence and stability ($k_s < 0$), $\hat{\nabla}(j)$ = estimated gradient vector of ε^2 with respect to W . Where the difference between the desired response and the output response forms the error signal $\varepsilon(j) = d(j) - W^T X(j)$. W^T = transpose of the weight vector and $W^T X(j)$ is output at j th sampling instant

$$\begin{aligned}\hat{\nabla}(j) &= \nabla \left[\varepsilon^2(j) \right] = 2 * \varepsilon(j) * \nabla(\varepsilon(j)) \\ \nabla(\varepsilon(j)) &= \nabla [d(j) - W^T X(j)] = -X(j) \\ \hat{\nabla}(j) &= -2 * \varepsilon(j) * X(j)\end{aligned}$$

using the gradient estimation formula the weight iteration rule becomes

$$W(j+1) = W(j) - 2k_s \varepsilon(j) X(j) \quad (7)$$

i.e., the next weight vector is obtained by adding to the present weight vector the input vector scaled by the value of the error. The LMS algorithm is directly usable as a weight adaptation formula for digital systems.

Computer simulations are concluded to ascertain the performance of the proposed LMS adaptive array. The array employed is a 21 element dipole of length $\lambda/2$ and spacing $\lambda/2$. All elements are assumed to be identical and Omni-directional with a unit gain. Here $\phi = 0$ and θ is varying desired signal coming from -20° to 20° . Jammers (4 Jammers) are radiating interference from 45° , 165° , -50° and -160° .

8. CONCLUSIONS

Techniques for interference suppression and anti-jamming which either dictate certain aspects of, or at least are relevant for, GPS receiver design include the following: (1) adaptive antenna arrays and associated electronics, (2) adaptive digital filters, (3) the addition of sensors (including inertial sensors) and associated hardware, and (2) processor architecture modifications necessitated for signal processing techniques; and adaptive narrowband filters. Adaptive array processing techniques attempt to remedy the drawbacks of the antenna switching concept by utilizing arrays of small antenna elements rather than full directional apertures. This makes installation on even small weapon platforms practical. Furthermore, implementation of phased-array processing is the key to significant multiple jammer suppression that can substantially restore GPS receiver operation in extensive jamming environments. The choice of one algorithm over another is determined by various factors. We are steering the beam in a specific direction. Our objective is fulfilled which reproduce the desired signal accurately as possible. We want the array processor to be very sensitive to the DOA of the signal. Here the issues of concern include (a) the number of operations (i.e., multiplications, divisions, and additions/subtractions) required making one complete iteration of the algorithm, (b) the size of memory locations required to store the data and the program, and (c) the investment required to program the algorithm on a computer or a DSP processor. So we can conclude that this onboard Adaptive antenna processing efficiently mitigate the intentional interference from jammers and keep military GPS receiver accurate.

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